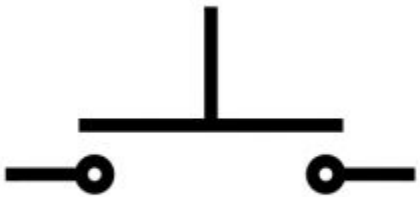
Digital Electronics Interfacing

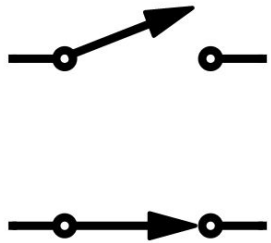
Lesson 5

Circuit Switches-Momentary and SPST(single pole single throw switch)

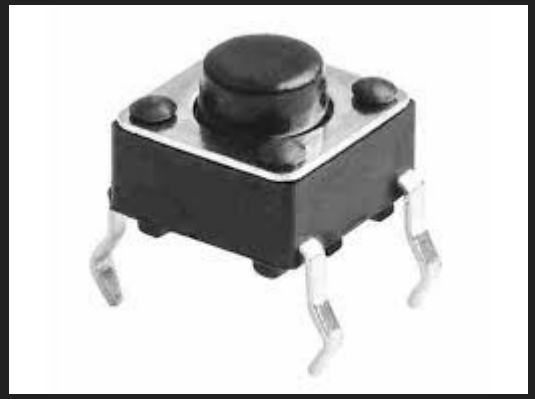
Push Button Switch

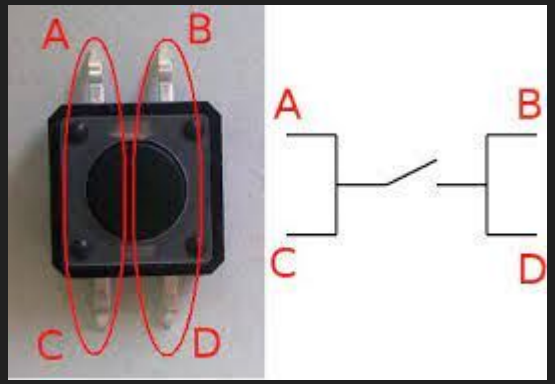


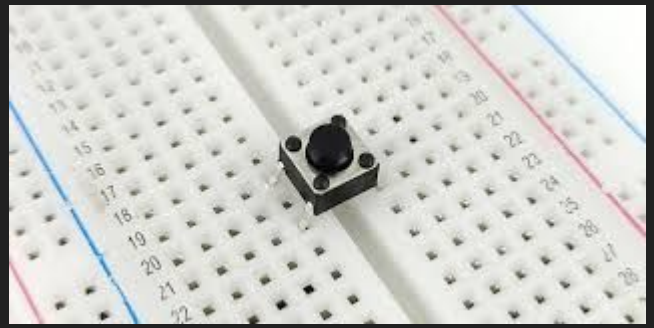
SPST

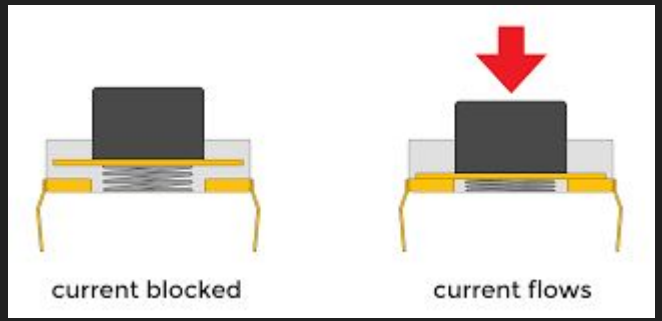


Breadboard Push Button

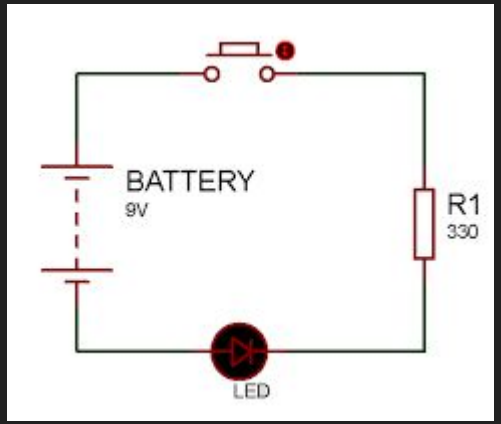








Schematic Circuit



Single Pole Single Throw Switches in TinkerCad

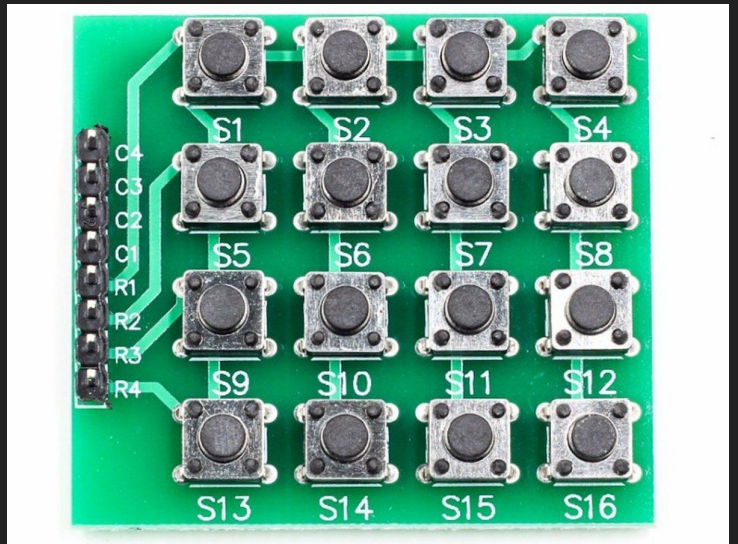




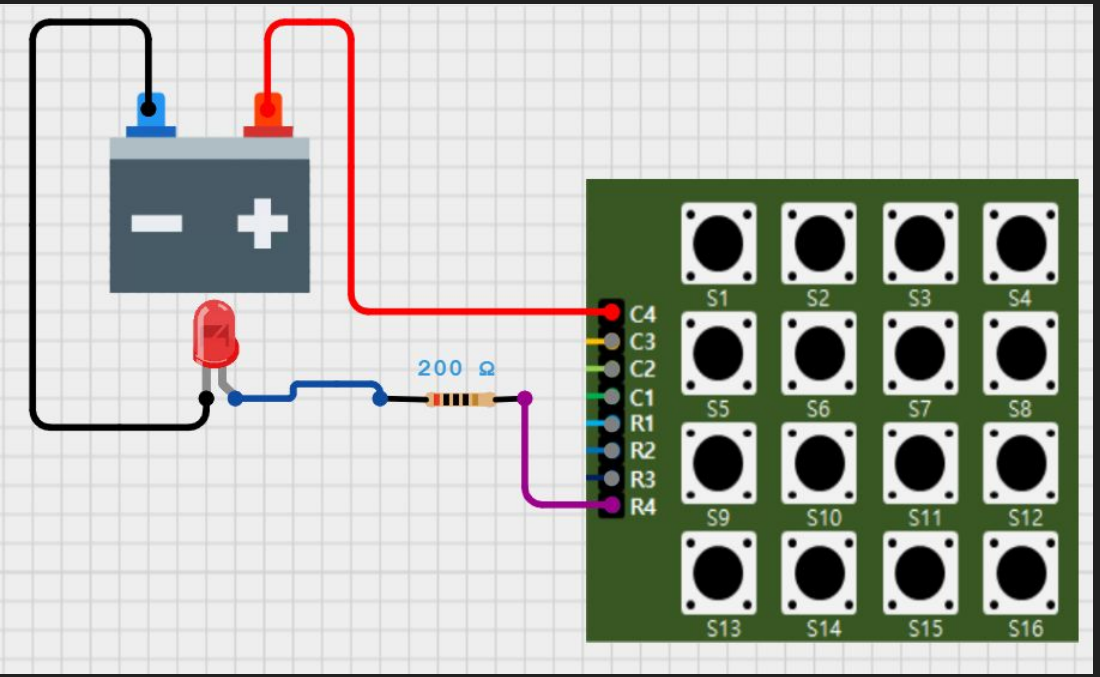
4x4 Push Button Matrix

This doesn’t exist in the tinkercad components list and therefore can not be simulated there.

This component can be used to get access to up to 16 buttons. The buttons are arranged in 4 columns and 4 rows. Button S1 is located at C1 and R1 and S16 is located at C4 and R4. You can access the button through the pins labeled C4 to C1 (columns) and R1 to R4 (rows).

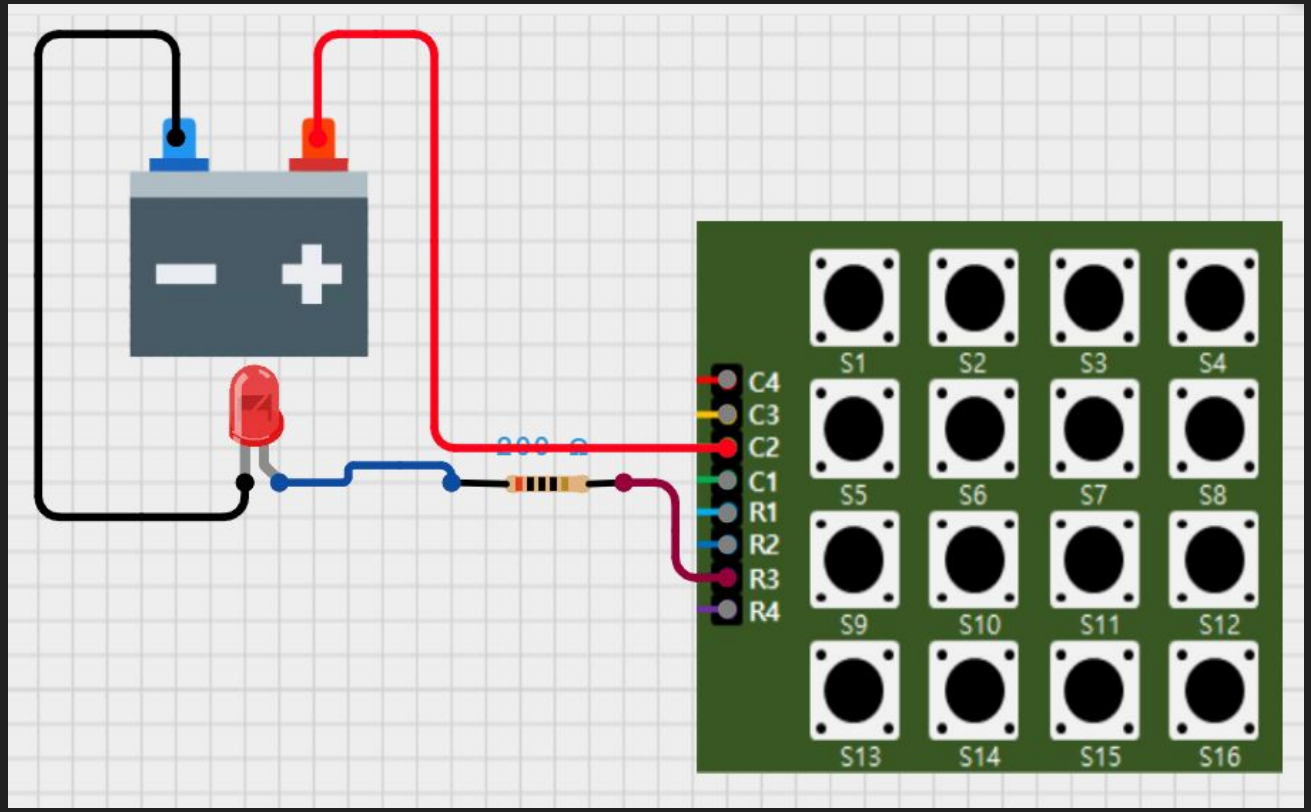


Controlling an LED using Button S16



In this image the LED will turn on when button S16 is pressed. The path between pin C4 and R4 is closed when this button is pressed, thereby completing the circuit and allowing current to flow from power, through the resistor, through the LED and back to ground.

Controlling an LED using Button S10



Jumper Wires

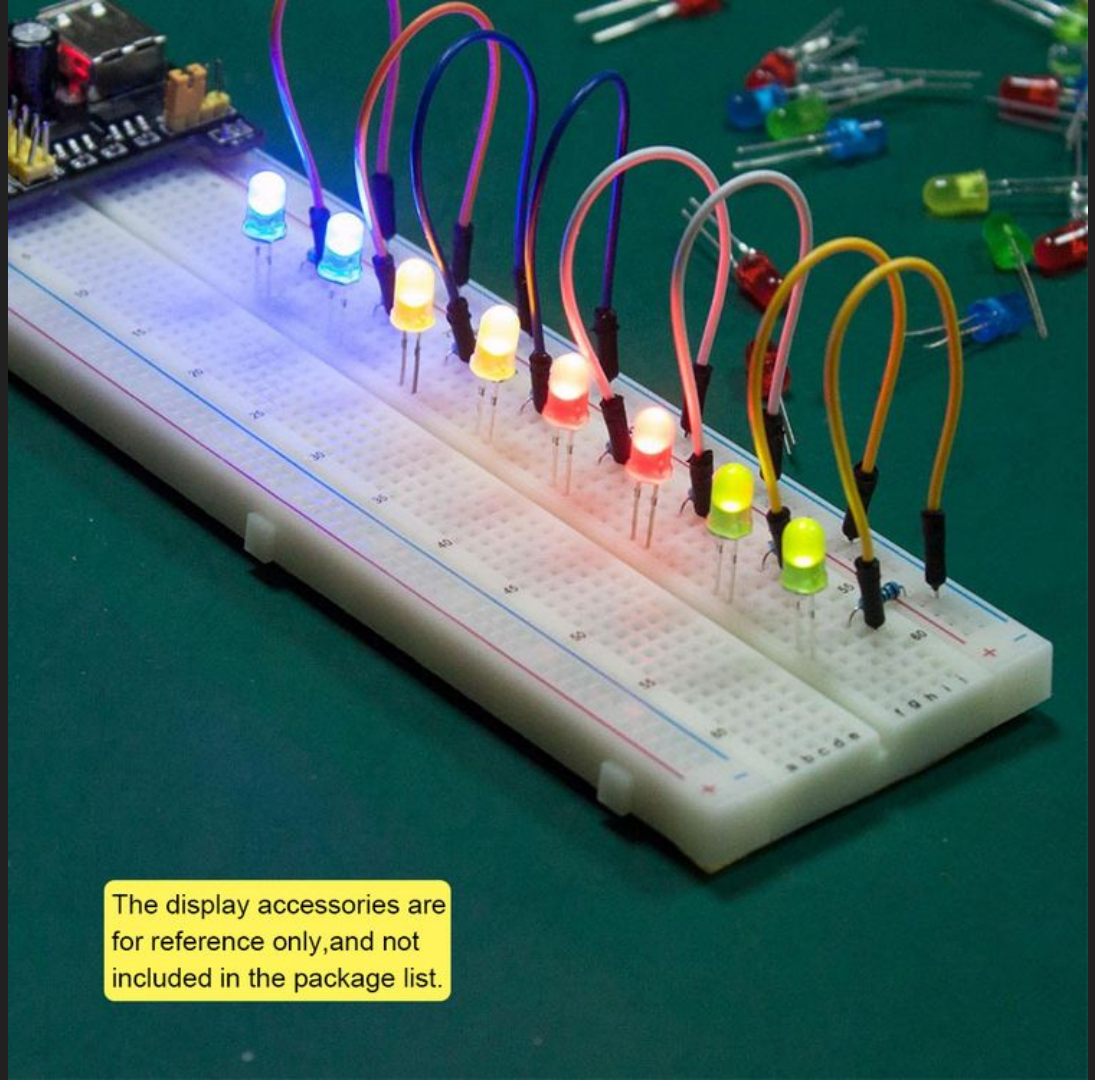
Premade wires with Dupont connectors

Commonly come as F-M,F-F,M-M

Common lengths are 10,20,30 and 40 cm

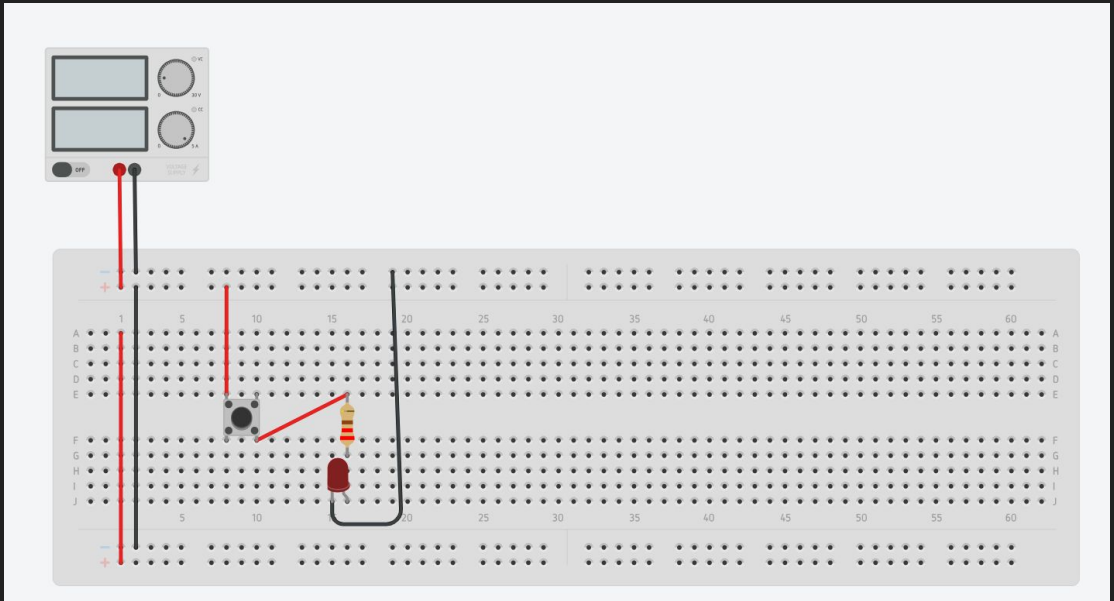
Used to connect different parts of a circuit in a breadboard.




The display accessories are for reference only,and not included in the package list.

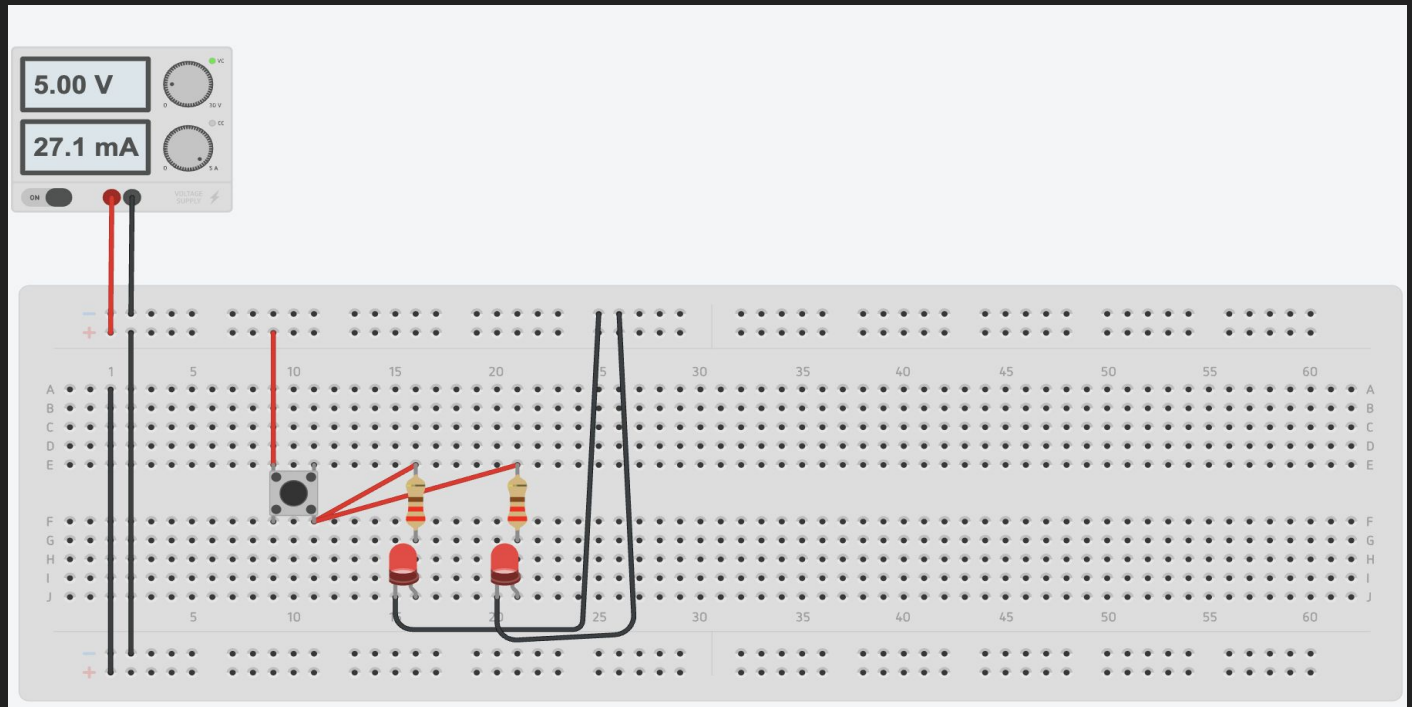
Questions and Exercises

1. For each of the following circuits embed a screenshot of the live and schematic view. Create a circuit where a LED is turned on and off by pressing a push button.



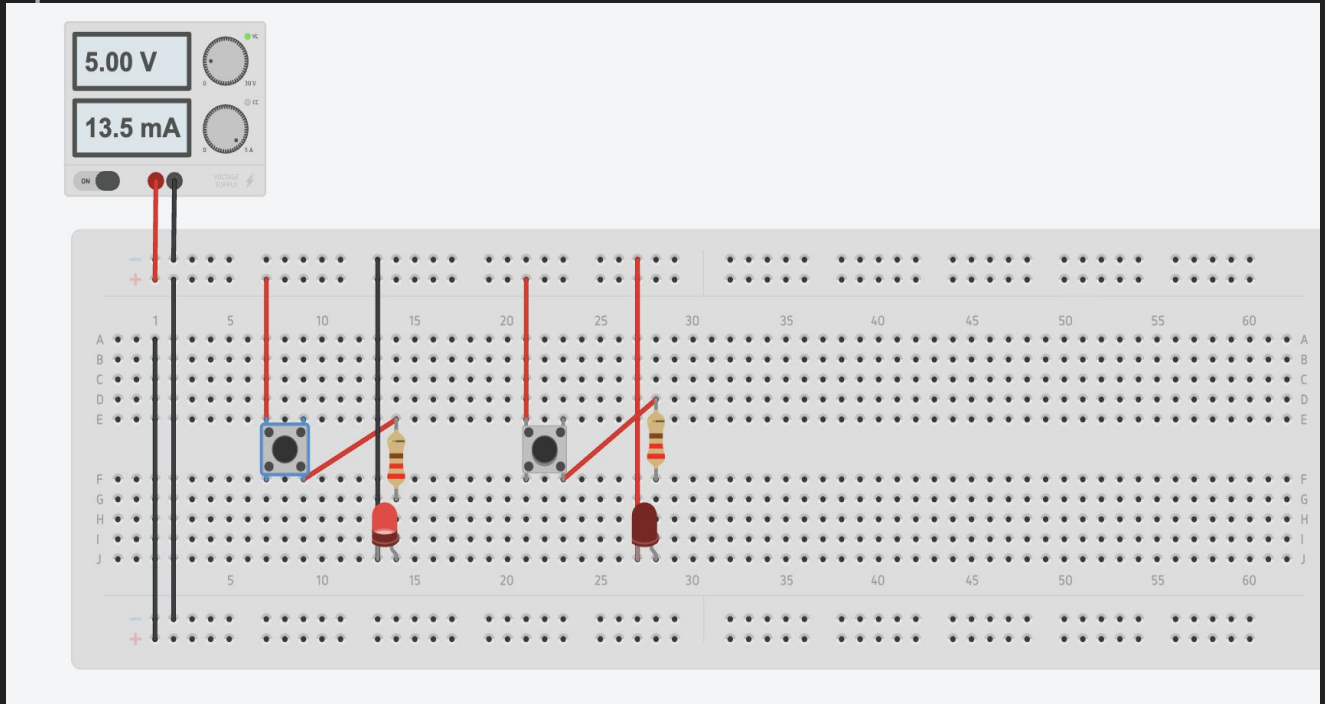
Questions and Exercises

2. Create a circuit where 2 LEDs are turned on and off by pressing a push button.



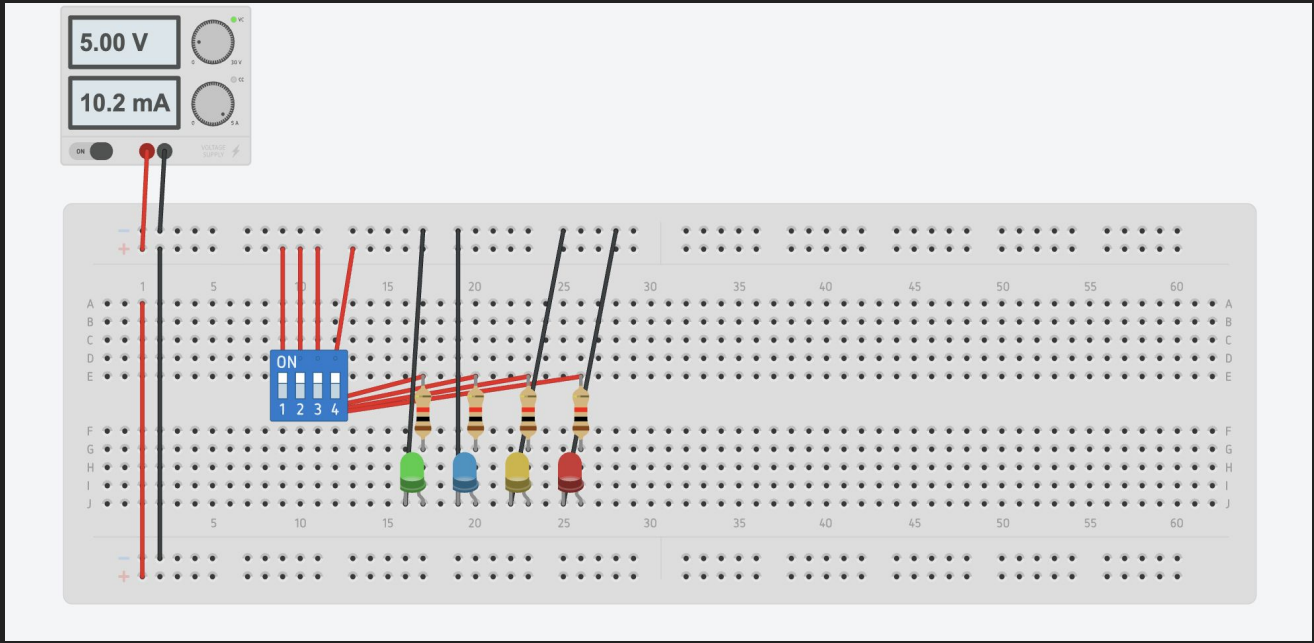
Questions and Exercises

3. Create a circuit where 2 LEDs are turned on and off by pressing two separate push buttons.



Questions and Exercises

4. Create a circuit where 4 LEDs are controlled by a 4 SPST switch in Tinkercad.



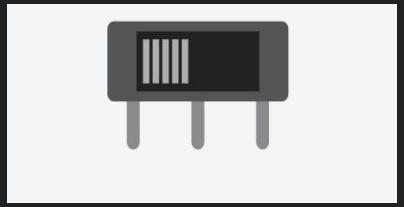
Questions and Exercises

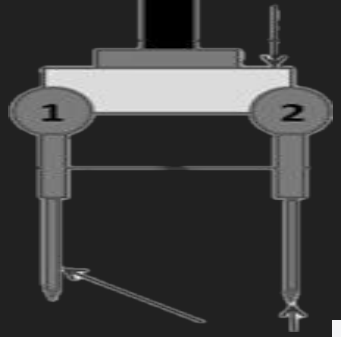
5. Do some research on a slide switch. Take a screenshot of the slide switch used in Tinkercad and embed it here.

Find an image on the internet of the internal design of a slide switch and embed it here.

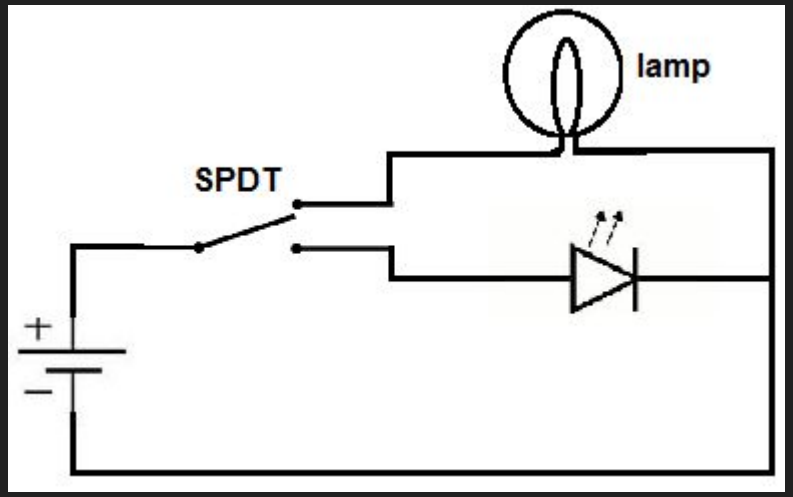
Slide switches are referred to as SPDT switches. What does this acronym stand for. Find and embed a schematic of this type of switch.

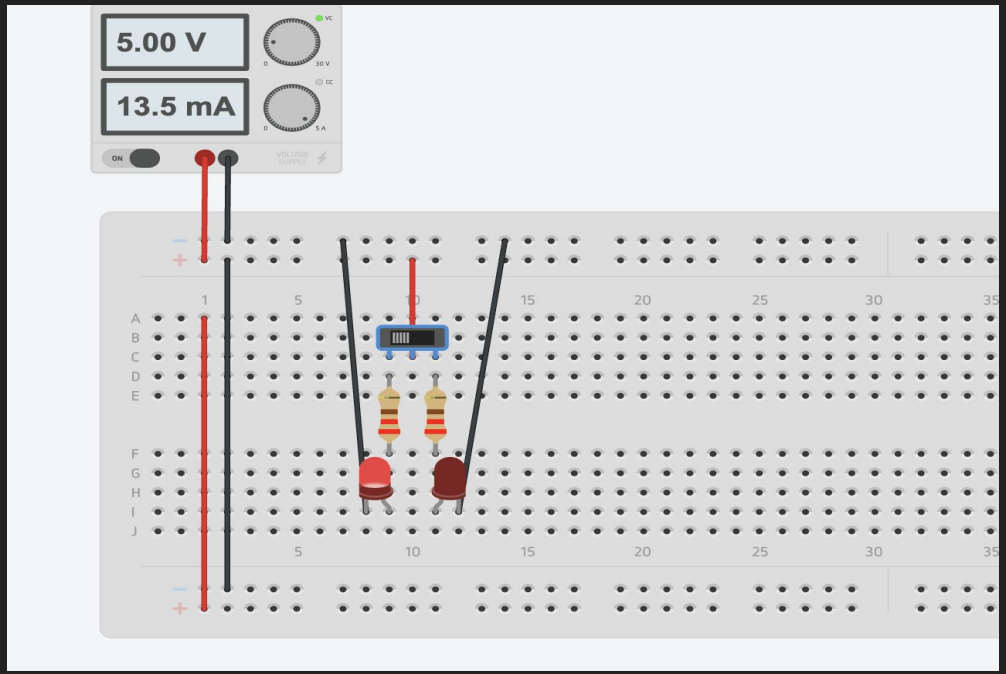
Design a tinkercad and live circuit using a slide switch to turn on and off two LEDs.





SPDT = Single Pole, Double Throw





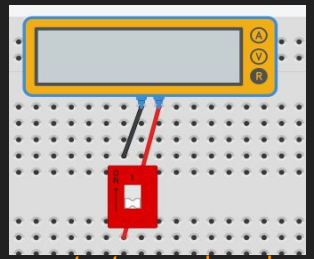
Questions and Exercises

6. Create a tinkercad circuit with DPST switch. What does DPST stand for? Find and embed a schematic of a DPST switch here:

Double Pole, Single Throw



Add a multimeter to your circuit. Set it to measure resis cross one set of poles in the switch (see screenshot):



Write down here what values you get when the switch is in each of

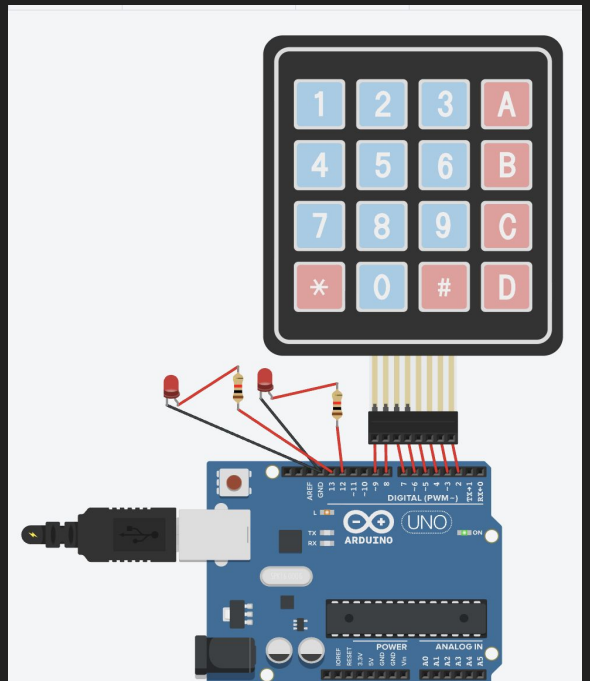
its two positions. Explain your results.

Across one pole: When the switch is OFF, the circuit is open and the resistance is infinite. When the switch is ON, the contacts are closed and the resistance is close to 0 Ω.

Now place the probes diagonally across each pole and repeat the process.

Explain your results.**Diagonal measurement:**The resistance is infinite in both ON and OFF positions because the two poles are electrically isolated.

7. Create a live circuit build using the matrix button component where you turn on and off 2 LEDs using the diagonal set of buttons S1 and S16. Embed a picture of your live circuit below.



Switch Assignment /10 TI

Create a live circuit with a single push button, 2 slide switches, 4 x SPST switch and a matrix button switch. A single LED will turn on only when the following occurs:

-the push button is pressed

-one slide switch is switched to the left and the other to the right

-the first and third 4 x SPST slide switches are on

-two opposite diagonal buttons on the matrix are pressed

You can test for most of this, minus the matrix, on tinkercad and then once everything works, add in the matrix. If all you can do is everything minus the matrix, you will be awarded 8 marks.

https://www.tinkercad.com/things/hSahbBz2dNO-bodacious-wolt?sharecode=j7_7a44MGEow90iD7XNYl-o7XqXmE4vNYlYL6SrStA0